Addition in $\mathbb{N}$

1. $7 + 4$
2. $23 + 15$
3. $148 + 32$
4. $467 + 285$
5. $1,250 + 4,832$
6. $56 + 189 + 43$
7. $342 + 1,056 + 89$
8. $2,415 + 3,890 + 764$
9. A library has $1,245$ fiction books, 856 non-fiction books, and 340 reference books. What is the total number of books in the library?
10. A manufacturing plant produced $4,280$ components in January, $5,125$ components in February, and $3,950$ components in March. How many total components were produced over the three-month period?

Solutions

1. **11**
   Explanation: Basic single-digit addition: $7 + 4 = 11$.

2. **38**
   Explanation: Align columns by place value:

$$\begin{array}{r} 2\,3 \\ +1\,5 \\ \hline \mathbf{3\,8} \end{array}$$

   Add the ones digits ($3 + 5 = 8$) and the tens digits ($2 + 1 = 3$), resulting in 38.

3. **180**
   Explanation: Align columns by place value:

$$\begin{array}{r} {\scriptstyle 1} \\ 1\,4\,8 \\ +\,3\,2 \\ \hline \mathbf{1\,8\,0} \end{array}$$

   Ones: $8 + 2 = 10$ (write 0, carry 1 to tens). Tens: $4 + 3 + 1 = 8$. Hundreds: 1. Total: 180.

4. **752**
   Explanation: Column addition with regrouping:

$$\begin{array}{r} {\scriptstyle 1\ 1} \\ 4\,6\,7 \\ +2\,8\,5 \\ \hline \mathbf{7\,5\,2} \end{array}$$

   Ones: $7+5 = 12$ (write 2, carry 1 to tens). Tens: $6+8+1 = 15$ (write 5, carry 1 to hundreds). Hundreds: $4 + 2 + 1 = 7$. Total: 752.

5. **6,082**
   Explanation: Multi-digit column layout:

$$\begin{array}{r} {\scriptstyle 1} \\ 1\,2\,5\,0 \\ +4\,8\,3\,2 \\ \hline \mathbf{6\,0\,8\,2} \end{array}$$

   Ones: $0 + 2 = 2$. Tens: $5 + 3 = 8$. Hundreds: $2 + 8 = 10$ (write 0, carry 1 to thousands). Thousands: $1 + 4 + 1 = 6$. Total: $6,082$.

6. **288**
   Explanation: Aligning multiple terms vertically:

$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ 5\,6 \\ 1\,8\,9 \\ +\,4\,3 \\ \hline \mathbf{2\,8\,8} \end{array}$$

Ones column: $6 + 9 + 3 = 18$ (write 8, carry 1 to tens). Tens column: $5 + 8 + 4 + 1 = 18$ (write 8, carry 1 to hundreds). Hundreds column: $1 + 1 = 2$. Total: 288.

7. **1,487**
Explanation: Aligning terms by place value:

$$\begin{array}{r} {\scriptstyle 1\ 1\ \ } \\ 3\,4\,2 \\ 1\,0\,5\,6 \\ +\quad 8\,9 \\ \hline \mathbf{1\,4\,8\,7} \end{array}$$

Ones: $2 + 6 + 9 = 17$ (write 7, carry 1 to tens). Tens: $4 + 5 + 8 + 1 = 18$ (write 8, carry 1 to hundreds). Hundreds: $3 + 0 + 1 = 4$. Thousands: 1. Total: $1,487$.

8. **7,069**
Explanation: Standard column layout for three numbers:

$$\begin{array}{r} {\scriptstyle 2\ 1\ \ } \\ 2\,4\,1\,5 \\ 3\,8\,9\,0 \\ +\,7\,6\,4 \\ \hline \mathbf{7\,0\,6\,9} \end{array}$$

Ones: $5 + 0 + 4 = 9$. Tens: $1 + 9 + 6 = 16$ (write 6, carry 1 to hundreds). Hundreds: $4 + 8 + 7 + 1 = 20$ (write 0, carry 2 to thousands). Thousands: $2 + 3 + 2 = 7$. Total: $7,069$.

9. **2,441 books**
Explanation: Sum all categories vertically:

$$\begin{array}{r} {\scriptstyle 1\ 1\ 1\ \ } \\ 1\,2\,4\,5 \\ 8\,5\,6 \\ +\,3\,4\,0 \\ \hline \mathbf{2\,4\,4\,1} \end{array}$$

Ones: $5 + 6 + 0 = 11$ (write 1, carry 1 to tens). Tens: $4 + 5 + 4 + 1 = 14$ (write 4, carry 1 to hundreds). Hundreds: $2 + 8 + 3 + 1 = 14$ (write 4, carry 1 to thousands). Thousands: $1 + 1 = 2$. Total: $2,441$ books.

10. **13,355 components**
Explanation: Combine monthly production figures vertically:

$$\begin{array}{r} {\scriptstyle 1\ 1\ 1\ \ } \\ 4\,2\,8\,0 \\ 5\,1\,2\,5 \\ +\,3\,9\,5\,0 \\ \hline \mathbf{1\,3\,3\,5\,5} \end{array}$$

Ones: $0 + 5 + 0 = 5$. Tens: $8 + 2 + 5 = 15$ (write 5, carry 1 to hundreds). Hundreds: $2 + 1 + 9 + 1 = 13$ (write 3, carry 1 to thousands). Thousands: $4 + 5 + 3 + 1 = 13$ (write 3, carry 1 to ten-thousands). Total: $13,355$ components.